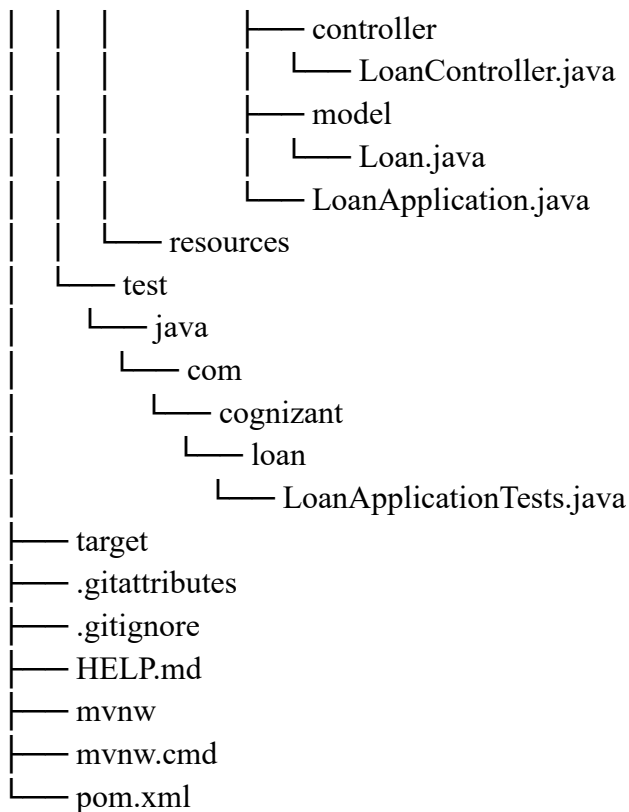


Module 8: Microservices with Spring Boot 3 and Spring Cloud

Creating Microservices for Account and Loan

The Project Directory:

```
microservices
├── account
│   ├── .mvn
│   ├── src
│   │   ├── main
│   │   │   ├── java
│   │   │   │   ├── com
│   │   │   │   │   ├── cognizant
│   │   │   │   │   │   ├── account
│   │   │   │   │   │   │   ├── controller
│   │   │   │   │   │   │   │   ├── AccountController.java
│   │   │   │   │   │   │   │   ├── model
│   │   │   │   │   │   │   │   │   ├── Account.java
│   │   │   │   │   │   │   │   │   AccountApplication.java
│   │   │   │   │   │   │   │   │   resources
│   │   │   │   │   │   │   │   │   test
│   │   │   │   │   │   │   │   │   │   ├── java
│   │   │   │   │   │   │   │   │   │   │   ├── com
│   │   │   │   │   │   │   │   │   │   │   ├── cognizant
│   │   │   │   │   │   │   │   │   │   │   ├── account
│   │   │   │   │   │   │   │   │   │   │   │   AccountApplicationTests.java
│   │   │   │   │   │   │   │   │   │   │   │   target
│   │   │   │   │   │   │   │   │   │   │   │   .gitattributes
│   │   │   │   │   │   │   │   │   │   │   │   .gitignore
│   │   │   │   │   │   │   │   │   │   │   │   HELP.md
│   │   │   │   │   │   │   │   │   │   │   │   mvnw
│   │   │   │   │   │   │   │   │   │   │   │   mvnw.cmd
│   │   │   │   │   │   │   │   │   │   │   │   pom.xml
│   │   │   │   │   │   │   │   │   │   │   │   loan
│   │   │   │   │   │   │   │   │   │   │   │   ├── .mvn
│   │   │   │   │   │   │   │   │   │   │   │   ├── src
│   │   │   │   │   │   │   │   │   │   │   │   │   ├── main
│   │   │   │   │   │   │   │   │   │   │   │   │   │   ├── java
│   │   │   │   │   │   │   │   │   │   │   │   │   │   ├── com
│   │   │   │   │   │   │   │   │   │   │   │   │   │   ├── cognizant
│   │   │   │   │   │   │   │   │   │   │   │   │   │   │   loan
```



Account

Codes:

AccountController.java

```
package com.cognizant.account.controller;
import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.PathVariable;
import org.springframework.web.bind.annotation.RequestMapping;
import org.springframework.web.bind.annotation.RestController;
import com.cognizant.account.model.Account;
@RestController
@RequestMapping("/accounts")
public class AccountController {
    @GetMapping("/{number}")
    public Account getAccount(@PathVariable String number) {
        return new Account(number, "savings", 234343);
    }
}
```

Account.java

```
package com.cognizant.account.model;
public class Account {
    private String number;
    private String type;
    private double balance;
    public Account() {
```

```

    }
    public Account(String number, String type, double balance) {
        this.number = number;
        this.type = type;
        this.balance = balance;
    }
    public String getNumber() {
        return number;
    }
    public void setNumber(String number) {
        this.number = number;
    }
    public String getType() {
        return type;
    }
    public void setType(String type) {
        this.type = type;
    }
    public double getBalance() {
        return balance;
    }
    public void setBalance(double balance) {
        this.balance = balance;
    }
}

```

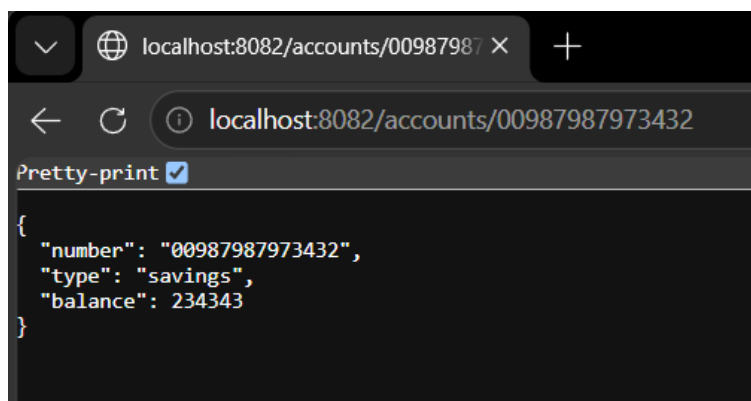
AccountApplication.java

```

package com.cognizant.account;
import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;
@SpringBootApplication
public class AccountApplication {
    public static void main(String[] args) {
        SpringApplication.run(AccountApplication.class, args);
    }
}

```

Output:



Loans

Codes:

LoanController.java

```
package com.cognizant.loan.controller;
import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.PathVariable;
import org.springframework.web.bind.annotation.RequestMapping;
import org.springframework.web.bind.annotation.RestController;
import com.cognizant.loan.model.Loan;
@RestController
@RequestMapping("/loans")
public class LoanController {
    @GetMapping("/{number}")
    public Loan getLoan(@PathVariable String number) {
        return new Loan(number, "car", 400000, 3258, 18);
    }
}
```

Loan.java

```
package com.cognizant.loan.model;
public class Loan {
    private String number;
    private String type;
    private double loan;
    private double emi;
    private int tenure;
    public Loan() {
    }
    public Loan(String number, String type, double loan, double emi, int
tenure) {
        this.number = number;
        this.type = type;
        this.loan = loan;
        this.emi = emi;
        this.tenure = tenure;
    }
    public String getNumber() {
        return number;
    }
    public void setNumber(String number) {
        this.number = number;
    }
    public String getType() {
        return type;
    }
    public void setType(String type) {
        this.type = type;
    }
}
```

```

    }
    public double getLoan() {
        return loan;
    }
    public void setLoan(double loan) {
        this.loan = loan;
    }
    public double getEmi() {
        return emi;
    }
    public void setEmi(double emi) {
        this.emi = emi;
    }
    public int getTenure() {
        return tenure;
    }
    public void setTenure(int tenure) {
        this.tenure = tenure;
    }
}

```

LoanApplication.java

```

package com.cognizant.loan;
import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;
@SpringBootApplication
public class LoanApplication {
    public static void main(String[] args) {
        SpringApplication.run(LoanApplication.class, args);
    }
}

```

Output:

